Geometry Relative to a Cut: Selection, Generation and Modular Inseparability

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Abstract

Can a geometric world and the structure of a perspective be specified independently? Standard half-sided modular inclusions give an exact setting in which to ask this question. The triple of two algebras and a vacuum determines a strongly additive chiral net, and the net determines the triple (Guido–Longo–Wiesbrock). Their geometric and modular data cannot be varied independently, although this symmetric correspondence establishes neither ontological priority nor experience.

We construct the inclusion rather than assuming it for a class of local seeds. In the vacuum representation of Vir_c, let $f(x) = xg(\log x)$, where g is a positive smooth bump, strictly increasing near its left endpoint and strictly decreasing near its right endpoint. The past modular orbit of the bounded observable $O_f = (2/\pi)\arctan T(f)$ generates a proper standard Cut. Its algebra contains a full tail half-line and lies inside a proper half-line of the original net. The proof uses geometric conjugation, smooth closure and positive translates of a localized bracket kernel; no independent forward-translation permission is needed. The full two-sided modular orbit consequently generates the entire Virasoro half-line algebra. For arbitrary nonzero stress seeds the earlier dilation-plus-forward-translation generation theorem remains valid.

In the SU(2) level-one net, two specified random rotations select exactly the $c = 1$ Virasoro subnet. Generation and noise selection therefore meet in one model, but at different levels: the selected subnet is not vacuum-cyclic on the ambient Hilbert space, whereas the Cut is standard in that subnet's own vacuum space. For local unitary excitations we also obtain the exact relative-entropy loss under Cut displacement, $2\pi aE$.

These results remove a separate Cut and a separate translation permission from the seed construction. The carrier, vacuum, modular evolution and ideal spectral and strong closures remain inputs. Finite-word tasks have a physical implementation in the neutral current model: local linear couplings, a retained helper and a paid autonomous clock approximate the required stress flows uniformly on bounded mean-energy inputs. Coherent control intervals and growing precision budgets are required. An exact Cut is not a finite observer, and modular flow is not thereby consciousness.

1 Introduction

Arguments for an explanatory gap between physics and experience [8] share a structure. A physical description is given: a space, a causal structure, matter fields in a state. Then one asks

why anything is experienced from a particular place within it. The gap appears because the description was complete before the question of a perspective arose. We call this the *independence assumption*: the physical structure can be specified without specifying a perspective, and the perspective is then added to, or located in, what was specified.

The assumption is an assumption about *specification*, and it can be examined in a limited way in an exact setting: ask whether the data that would constitute a perspective are an addition to the physical specification or are already fixed by it. An equivalence of descriptions does not settle which comes first, and it does not show that any algebra is a perspective; what it does settle is whether the perspective data can be varied independently of the physical ones. This paper examines that in the one setting where both sides are exact: von Neumann algebras of observables with a distinguished state.

Setting. *Geometry* enters through a conformal net on the circle: a family of algebras $\mathcal{A}(I)$ indexed by intervals, with the translations and dilations that move them. A *perspective* is represented by an accessible algebra A, a subalgebra of a global algebra M with a state ω. The state defines a one-parameter group of automorphisms of M, the modular flow σ_s. For the half-line algebra of a conformal net the modular flow is the dilation (Bisognano–Wichmann [5], Brunetti–Guido–Longo [3]), and in general the modular flow of an algebra in a state is the time that the state itself defines for that algebra (Connes–Rovelli [9]). We use the modular structure of an accessible algebra as the structural proxy for the temporal structure of a perspective. This is a proxy, not an identification; Section 8 says what it does and does not support.

Results.

(i) *Cuts and geometry* (Section 3). A proper half-sided modular inclusion $N \subset M$ forces a one-directional strict nesting of the modular orbit of N and is generated back into M by that orbit (Lemmas 3.1, 3.2); in the factor setting it lives only in type III_1 (Proposition 3.3). For any subalgebra $A \subset M$ whose past modular orbit is cyclic and proper, the Cut and its translations are reconstructed from (M, Ω, A), and the full net when the inclusion is also standard (Proposition 3.4). Geometry and the modular data of a standard Cut determine each other (Corollary 3.5), so they cannot be varied independently; the correspondence is symmetric and does not say which is prior.

(ii) *Selection* (Section 4). If a compact group acts on a net by internal symmetries and the unrecorded part of the environment applies it at random, the noise semigroup has the fixed-point algebra as its exact fixed algebra (Proposition 4.3). For SU(2) level one with $SO(3)$ this is the Virasoro net with $c = 1$ (Rehren), and two fixed rotations about non-parallel axes suffice to select it (Theorem 4.4). Several non-commuting noises select the intersection of their fixed algebras (Theorem 4.5). With the $\mathbb{Z}_2$ noise of the $U(1)$ current net the fixed algebra is strictly larger than the Virasoro net (Proposition 4.8).

(iii) *Generation* (Section 5). For positive endpoint-monotone logarithmic bumps, a single bounded stress seed and its past modular orbit generate a proper standard Cut (Theorem 5.7); the full modular orbit generates the Virasoro half-line algebra (Corollary 5.8). For every nonzero stress seed, without the bump restriction, dilations plus forward translations suffice (Theorem 5.1), while forward translations alone do not (Proposition 5.2). In SU(2) level one this generated algebra equals the noise-selected algebra (Theorem 5.4). The generated Cut is taken in its own vacuum representation, not in the larger ambient representation (Remark 5.5).

(iv) *Response* (Section 6). For local unitary excitations the relative entropy on the algebra of a displaced Cut is an exact expression in the null energy, and the loss under displacement is 2π times the null energy times the displacement (Theorem 6.1). Restriction to a smaller half-line therefore loses information even when the whole excitation remains accessible.

What enters the construction. The carrier algebra, vacuum and modular dynamics are specified. The seed construction then supplies a Cut and its reconstructed translations; it does not derive the carrier or its modular dynamics from a microscopic detector. Continuous parameters, spectral calculus and strong limits are used throughout. Section 7 connects these ideal structures to finite task permissions without identifying the two.

Modular decoherence and an ambient-algebra shell rule do not alone derive a proper half-sided inclusion; Appendix D records their boundaries. The constructive route here uses local stress fields and geometric covariance. The original question remains: which physical history produces a usable boundary, and what does that boundary contribute to a perspective?

2 Preliminaries

2.1 Modular theory and half-sided modular inclusions

Let M be a von Neumann algebra on a Hilbert space H and $\Omega \in H$ a vector that is cyclic and separating for M. Write Δ_M, J_M for the modular operator and conjugation of (M, Ω) and $\sigma_s^M(x) = \Delta_M^{is} x \Delta_M^{-is}$. Then $\sigma_s^M(M) = M$ and $\Delta_M^{is}\Omega = \Omega$.

Definition 2.1 (Half-sided modular inclusion)**.** Let $N \subset M$ be von Neumann algebras for which Ω is cyclic and separating. The inclusion is *+hsm* if $\sigma_s^M(N) \subset N$ for all $s \leq 0$ (equivalently $\Delta_M^{-it} N \Delta_M^{it} \subset N$ for $t \geq 0$). It is *standard* if Ω is also cyclic for $M \cap N'$. It is *proper* if $N \neq M$. We call a proper +hsm inclusion a *Cut*.

Theorem 2.2 (Wiesbrock; Araki–Zsidó [2, Thm. 2.1])**.** *Let $N \subset M$ be +hsm. Then $P := \overline{(\log \Delta_N - \log \Delta_M)/2\pi}$ is a positive self-adjoint operator, and with $U(s) = e^{isP}$:*

(1) $\Delta_M^{-it} U(s) \Delta_M^{it} = U(e^{2\pi t} s) = \Delta_N^{-it} U(s) \Delta_N^{it}$;

(2) $J_M U(s) J_M = U(-s) = J_N U(s) J_N$;

(3) $\Delta_N^{it} = U(1) \Delta_M^{it} U(1)^*$ *and* $J_N = U(1) J_M U(1)^*$;

(4) $N = U(1) M U(1)^*$;

(5) $U(s) M U(s)^* \subset M$ *for* $s \geq 0$, *and* $\mathrm{Ad}\, U(s)$, $s \geq 0$, *is a strongly continuous semigroup of endomorphisms of* M;

(6) $U(s)\Omega = \Omega$, *so the vacuum state is invariant under* $\mathrm{Ad}\, U(s)$.

Theorem 2.3 ([2, Thm. 2.2(4)], [14, Cor. 1.8])**.** *If $N \subset M$ is +hsm and p is the largest central projection of M with $Mp = Np$, then $M(1-p)$ and $N(1-p)$ are of type III$_1$ whenever $p \neq 1$. In particular, if M is a factor and $N \neq M$, both are type III$_1$ factors.*

Theorem 2.4 (Guido–Longo–Wiesbrock [14, Cor. 1.7–1.9])**.** *There is a one-to-one correspondence between isomorphism classes of* standard *+hsm inclusions $(N \subset M, \Omega)$ and isomorphism classes of strongly additive local conformal nets on S^1; under it $M = \mathcal{A}(0,\infty)$, $N = \mathcal{A}(1,\infty)$, Ω is the vacuum, U is the translation group, and Δ_M^{it} acts as dilation. For a conformal net that is not strongly additive, the hsm reconstruction returns the dual net $\mathcal{A}^d(a,b) = \mathcal{A}(a,\infty) \cap \mathcal{A}(b,\infty)'$, which equals $\mathcal{A}$ if and only if $\mathcal{A}$ is strongly additive.*

2.2 Virasoro and current nets

A conformal net $\mathcal{A}$ is *diffeomorphism covariant* with central charge $c > 0$ if the vacuum representation carries a positive-energy projective representation V of $\mathrm{Diff}_+(S^1)$, with $V(\varphi) \in \mathcal{A}(I)$ when φ is supported in I, and stress tensor T with $V(\mathrm{Exp}\, f) = e^{iT(f)}$ up to a phase (Goodman–Wallach [13], Carpi–Weiner [10]). Write $\mathrm{Vir}_c(I) = \{T(f) : \mathrm{supp} f \subset I\}''$. For real $f \in C_c^\infty$, $T(f)$ is essentially self-adjoint on finite-energy vectors, and if $f_n \to f$ in $\|f\|_{3/2} = \sum_n |\hat{f}_n|(1+|n|^{3/2})$ with supports in a fixed compact set, then $T(f_n) \to T(f)$ in the strong resolvent sense [10, Thm. 4.4, Prop. 4.5]. For the half-line $M = \mathrm{Vir}_c(0,\infty)$ the modular group of the vacuum is the dilation group, $\sigma_s = \mathrm{Ad}\, V(\delta_{e^{-2\pi s}})$ with $\delta_\lambda(x) = \lambda x$ (Bisognano–Wichmann [5]; Brunetti–Guido–Longo [3]), and $T_a = \mathrm{Ad}\, U(a)$ with $U(a) = V(\tau_a)$ implements translations. The transformation law is $V(\varphi)T(x)V(\varphi)^* = \varphi'(x)^2 T(\varphi(x)) - \frac{c}{24\pi}\mathrm{Sch}(\varphi)(x)$. Virasoro nets with $c \le 1$ are strongly additive and for $c > 1$ they are not [7, §2.4]; the fixed-point subnet of a strongly additive net under a compact group is strongly additive [25].

2.3 Relative entropy of coherent states

Let $f \in C_c^\infty((a_0,\infty),\mathbb{R})$ with $a_0 > 0$, $\rho = \mathrm{Exp}\, f$, $U = V(\rho) = e^{iT(f)}$, $\omega_+(A) = \omega(U^*AU)$ and $\omega_-(A) = \omega(UAU^*)$. Put $w = \rho''/\rho'$. Hollands [15, Thm. 1] proves, for the half-line algebra $\mathrm{Vir}_c(0,\infty)$,

$$S(\omega\|\omega_+) = \frac{c}{24}\int_0^\infty x\,(\partial_x \log \rho'(x))^2\, dx. \tag{1}$$

3 Cuts and geometry

Let $(N \subset M, \Omega)$ be $+$hsm with translation group U from Theorem 2.2. Put $N_a = U(a)MU(a)^*$ for $a \ge 0$, so $N_0 = M$, $N_1 = N$.

Lemma 3.1 (The modular orbit of a Cut generates M)**.** $\bigvee_{t\in\mathbb{R}} \sigma_t^M(N) = M$.

Proof. By Theorem 2.2(1),(4) and $\sigma_t(M) = M$, $\sigma_t(N) = \Delta_M^{it}U(1)MU(1)^*\Delta_M^{-it} = U(e^{-2\pi t})MU(e^{-2\pi t})^* = N_{e^{-2\pi t}}$. For $m \in M$ put $m_a = U(a)mU(a)^* \in N_a$. As $t \to \infty$, $a = e^{-2\pi t} \to 0$ and $m_a \to m$ strongly, since U is strongly continuous. Each m_a lies in $\sigma_t(N)$ for $a = e^{-2\pi t}$, and the algebra generated by the $\sigma_t(N)$ is strongly closed, so it contains m. The other inclusion is trivial. □

Lemma 3.2 (One-directional strict nesting)**.** *If $N \subsetneq M$ is $+$hsm, then $\sigma_t(N) \subsetneq N$ for $t < 0$ and $\sigma_t(N) \not\subset N$ for $t > 0$.*

Proof. For $a \ge b \ge 0$, $N_a = U(b)[U(a-b)MU(a-b)^*]U(b)^* \subset U(b)MU(b)^* = N_b$ by Theorem 2.2(5), so $a \mapsto N_a$ is decreasing. Suppose $N_a = N_b$ for some $a > b \ge 0$. Conjugating by $U(b)^*$ gives $N_d = M$ with $d = a - b > 0$. Choose t with $e^{-2\pi t}d = 1$: then $N = N_1 = \sigma_t(N_d) = \sigma_t(M) = M$, a contradiction. Hence $a \mapsto N_a$ is strictly decreasing. Since $\sigma_t(N) = N_{e^{-2\pi t}}$, the claims follow. □

The nesting is a property of the subalgebras. It is not entropy production or an arrow of time: the modular group itself is invertible and acts by automorphisms. Lemma 3.2 says that a proper Cut cannot be stable in both directions.

Proposition 3.3 (A Cut lives in type III_1)**.** *If M is a factor and $N \subsetneq M$ is $+$hsm, then M and N are type III_1 factors. No proper $+$hsm inclusion exists in a type I factor.*

This is Theorem 2.3. It is a statement about the type of the algebras, not about observers: a finite-dimensional system has no exact Cut, but an observer of finite capacity can still couple to a system described by a type III_1 algebra. Finite instruments and exact Cuts therefore require different descriptions; Section 7 gives their task-level interface.

Proposition 3.4 (Accessible algebras and geometry)**.** *Let $A \subset M$ be a subalgebra and $N_A^- = \big(\bigcup_{s\le 0}\sigma_s(A)\big)'' \subset M$. Assume that Ω is cyclic for N_A^- and $N_A^- \neq M$. Then:*

(a) *$(N_A^- \subset M, \Omega)$ is a Cut, with translation group U_A and $N_A^- = U_A(1) M U_A(1)^*$;*

(b) *$\Delta^{it}_{N_A^-} = U_A(1)\Delta_M^{it}U_A(1)^*$, and U_A is recovered from the two modular operators as $P = \dfrac{\cdot}{(\log\Delta_{N_A^-} - \log\Delta_M)/2\pi}$;*

(c) *if the inclusion is standard, there is a unique strongly additive local conformal net $\mathcal{A}_A$ with $\mathcal{A}_A(0,\infty) = M$, $\mathcal{A}_A(1,\infty) = N_A^-$, translations U_A and dilations σ;*

(d) *$\bigvee_t \sigma_t(A) = M$;*

(e) *if $A \subset A' \subset M$ and both satisfy the hypotheses, then $N_A^- \subset N_{A'}^-$.*

Proof. $\sigma_s(N_A^-) \subset N_A^-$ for $s \le 0$ by construction, and Ω is separating for $N_A^- \subset M$, so (a) is Theorem 2.2(4), and (b) is (1),(3). (c) is Theorem 2.4. (d): $\bigvee_t \sigma_t(A) \supset \bigvee_t \sigma_t(N_A^-) = M$ by Lemma 3.1, and the left side is contained in M. (e) is monotonicity of the generated algebra. □

Corollary 3.5 (Geometry and modular data determine each other)**.** *Within the class of standard Cuts, the geometry (the isomorphism class of the strongly additive net) and the modular data (the isomorphism class of the triple $(N \subset M, \Omega)$) determine each other.*

This is Theorem 2.4. The input is the complete standard triple, not a bare flow: Δ_M and U together with M and Ω fix N and Δ_N, and conversely the two modular operators fix U. The triple replaces the list "net, vacuum, translations, dilations"; it does not explain where M, Ω or the Cut come from.

Remark 3.6 (Cuts on one orbit are isomorphic). The Cuts N_a and N_b of one net, $a, b > 0$, are exchanged by σ_t and give isomorphic triples. Distinguishing geometries therefore needs an invariant of the triple (for instance the spectrum of the conformal Hamiltonian of the reconstructed net), not merely a different subalgebra. Foundation I supplies one in this net: the vacuum character of the hull determines the stabilizer of the seed up to conjugacy within that model, so seeds with non-conjugate stabilizers give different vacuum characters and hence non-isomorphic triples [20]. That invariant classifies Foundation I's ambient-translation Cuts $U(1)\,\mathrm{Hull}(S)U(1)^* \subset \mathrm{Hull}(S)$. It does not classify the past-orbit inclusion $N_{O_f}^- \subset M$ by identifying its reconstructed U_A with the ambient U. Conjecture 5.10(b) concerns precisely whether that orbit is an ordinary translated half-line.

Remark 3.7 (Relative position of regions). For nested intervals $(x_2, x_3) \subset (x_1, x_4)$ the cross-ratio $b = \frac{(x_2-x_1)(x_4-x_3)}{(x_3-x_1)(x_4-x_2)} \in (0,1)$ is the unique modulus under $PSL(2,\mathbb{R})$; in the normal form the outer interval is $(0,\infty)$ and the inner one is $(b,1)$. Any statement about the relative position of nested regions in the reconstructed net is a statement about b.

4 Selection of the accessible algebra by noise

4.1 Setting

Assumption 4.1. $\mathcal{A}$ is an irreducible, local, Möbius covariant net on S^1 with vacuum (H, Ω). A compact group G acts by automorphisms α_g, preserving each $\mathcal{A}(I)$, implemented by a strongly

continuous unitary representation $u(g)$ with $u(g)\Omega = \Omega$ that commutes with the Möbius (and, when present, the diffeomorphism) representation. We write $\omega = \langle \Omega, \cdot\, \Omega\rangle$ and $M = \mathcal{A}(0,\infty)$.

For a closed subgroup $H \subset G$ put $Q_H(x) = \int_H \alpha_h(x)\, dh$ with normalised Haar measure.

Lemma 4.2. *Q_H is a normal, faithful, ω-preserving conditional expectation of M onto the fixed algebra M^H; it is M^H-bimodular and commutes with the modular group and with translations.*

Proof. For a vector functional $\rho = \langle \xi, \cdot\, \eta\rangle$ the map $h \mapsto \rho\circ\alpha_h$ is norm continuous in the predual, by strong continuity of u; every normal functional is a norm limit of finite sums of vector functionals, so the same holds for all normal ρ. The Bochner integral $\int_H \rho\circ\alpha_h\, dh$ is therefore a normal functional, and it equals $\rho\circ Q_H$; hence Q_H is normal. Haar invariance gives $Q_H^2 = Q_H$ and $\mathrm{Ran}\, Q_H = M^H$; bimodularity is $\alpha_h(n_1 x n_2) = n_1\alpha_h(x)n_2$ for $n_i \in M^H$. Since $\omega\circ\alpha_h = \omega$, $\omega\circ Q_H = \omega$, and if $Q_H(x^*x) = 0$ then $\omega(x^*x) = 0$, so $x = 0$ (faithfulness). Commutation with σ and translations follows because $u(g)$ commutes with their unitary groups. □

4.2 Noise semigroups

Fix $\kappa > 0$ and put $D_t^H = e^{-\kappa t}\,\mathrm{id} + (1 - e^{-\kappa t})Q_H$. This is the average channel of a Poisson process of rate κ in which each jump applies α_h with h Haar-distributed in H: the probability of no jump in time t is $e^{-\kappa t}$, and a product of at least one Haar element is again Haar.

Proposition 4.3. *D_t^H is a norm-continuous semigroup of normal, unital, completely positive, ω-preserving maps on M, and*

(1) $\mathrm{Fix}(D^H) = M^H$*; $D_t^H - Q_H = e^{-\kappa t}(\mathrm{id} - Q_H)$, so $\|D_t^H - Q_H\| \le 2e^{-\kappa t}$;*

(2) *$H_1 \subset H_2 \Rightarrow M^{H_2} \subset M^{H_1}$; κ changes only the rate of approach;*

(3) *if $M^H \neq M$ and $0 \neq y = y^* \in \ker Q_H$, then for $0 < t < \infty$ and $q = e^{-\kappa t}$ we have $D_t(y^2) - D_t(y)^2 = (1-q)Q_H(y^2) + (q - q^2)y^2 \ge 0$ and $\neq 0$, so D_t is a non-multiplicative channel although it preserves ω.*

Proof. The semigroup law and complete positivity are those of a convex combination of id and Q_H with $Q_H^2 = Q_H$. (1): if $D_t(x) = x$ for some $t > 0$ then $(1 - e^{-\kappa t})(x - Q_H x) = 0$. (2) is clear. (3): choose $x = x^* \notin M^H$ and $y = x - Q_H x$; then $Q_H y = 0$, $D_t(y) = qy$, and $D_t(y^2) = qy^2 + (1-q)Q_H(y^2)$. Since Q_H is faithful, $Q_H(y^2) \neq 0$. □

Theorem 4.4 (Two fixed rotations suffice)**.** *Let $G = SO(3)$ act as in Assumption 4.1. Let $g_1, \dots, g_m \in SO(3)$ generate a subgroup whose closure is $SO(3)$, let $p_j > 0$, $\sum_j p_j = 1$, and put $R = \sum_j p_j \alpha_{g_j}$, $\tilde{D}_t = \exp[\kappa t(R - \mathrm{id})]$. Then $\tilde{D}$ is a norm-continuous semigroup of normal, unital, completely positive, ω-preserving maps with $\mathrm{Fix}(\tilde{D}) = M^{SO(3)}$. The hypothesis holds, in particular, (i) for $m = 2$ when g_1, g_2 are rotations about two non-parallel axes by angles whose ratios to 2π are irrational (any $p_1 \in (0,1)$), and (ii) for the symmetric four-operation choice $R = \frac{1}{4}(\alpha_{g_1} + \alpha_{g_1^{-1}} + \alpha_{g_2} + \alpha_{g_2^{-1}})$ built from such a pair.*

Proof. The generator is bounded, so $\mathrm{Fix}(\tilde{D}) = \mathrm{Fix}(R)$. Write α_j for the automorphisms and u_j for their implementers, which fix Ω. Then $\|(\alpha_j(x) - x)\Omega\|^2 = 2\|x\Omega\|^2 - 2\mathrm{Re}\langle \alpha_j(x)\Omega, x\Omega\rangle$, so $\sum_j p_j\|(\alpha_j(x) - x)\Omega\|^2 = 2\|x\Omega\|^2 - 2\mathrm{Re}\langle R(x)\Omega, x\Omega\rangle$. If $R(x) = x$, the left side vanishes, so $\alpha_j(x)\Omega = x\Omega$ for each j, and $\alpha_j(x) = x$ because Ω is separating. The set $\{g : \alpha_g(x) = x\} = \{g : u(g)x\Omega = x\Omega\}$ is a closed subgroup (strong continuity of u) that contains every g_j, hence the closure of the group they generate. For (i): each g_j generates a dense subgroup of the circle

T_j of rotations about its axis n_j, so the closure H of $\langle g_1, g_2\rangle$ is a closed, hence Lie, subgroup containing T_1 and T_2. Its Lie algebra contains $n_1, n_2 \in \mathfrak{so}(3) \cong \mathbb{R}^3$ (bracket = cross product), hence $n_1 \times n_2 \neq 0$; the three vectors span $\mathbb{R}^3$, so H contains the identity component of $SO(3)$, which is $SO(3)$. For (ii) the generated group is the same. Then x is fixed by all of $SO(3)$; the converse inclusion is clear. □

Two is not claimed to be the minimal number of operations in every mechanism.

Theorem 4.5 (Non-commuting noises select the intersection)**.** *Let $H_1, \dots, H_m \subset G$ be closed subgroups, $\kappa_i > 0$, $L = \sum_i \kappa_i (Q_{H_i} - \mathrm{id})$, $E_t = e^{tL}$. The Q_{H_i} need not commute. Then E_t is a norm-continuous semigroup of normal, unital, completely positive, ω-preserving maps, with*

$$\mathrm{Fix}(E) = \bigcap_i M^{H_i} = M^{\overline{\langle H_1,\dots,H_m\rangle}}, \qquad -\langle x\Omega, Lx\Omega\rangle = \sum_i \kappa_i \|(x - Q_{H_i}x)\Omega\|^2,$$

and E_t converges strongly on $L^2(M,\omega)$ to the Haar average over the closed group generated.

Proof. With $\lambda = \sum \kappa_i$ and $\bar{R} = \sum \kappa_i Q_{H_i}/\lambda$, $E_t = e^{-\lambda t}\sum_n (\lambda t)^n \bar{R}^n/n!$ is a Poisson mixture of powers of a state-preserving unital completely positive map. On the GNS space each Q_{H_i} is the orthogonal projection $P_i = \int_{H_i} u(h)\, dh$; hence L is bounded, self-adjoint and nonpositive there, and the displayed identity holds. $Lx = 0$ forces $Q_{H_i}x = x$ for every i, because Ω is separating. The spectral theorem gives strong convergence to the projection onto $\ker L = \bigcap_i \mathrm{Ran} P_i$, which is the Haar average over the closed group generated. □

No uniform spectral gap or operator-norm rate is claimed in infinite dimension.

4.3 SU(2) level one and the U(1) contrast

Let $\mathcal{A}_{SU(2)_1}$ be the vacuum net of the loop group $LSU(2)$ at level one; $SO(3)$ acts properly by the adjoint action, and Vir_1 denotes the Virasoro subnet of the Sugawara stress tensor, $c = 1$.

Theorem 4.6 (Rehren [17])**.** $\mathcal{A}_{SU(2)_1}(I)^{SO(3)} = \mathrm{Vir}_1(I)$ *for every interval I.*

The proof in [17] decomposes the vacuum representation by isospin: its character is $\sum_j \chi_j(z)\, t^{j^2}(1 - t^{2j+1})\prod_n (1 - t^n)^{-1}$, the $SO(3)$-invariant part is the $j = 0$ term, which is the character of the irreducible Virasoro vacuum representation at $c = 1$ (Segal [18], Kac–Raina [16]), so the invariant vacuum space equals the Virasoro vacuum space. Haag duality of the Virasoro net [6] and locality then give the chain $\mathrm{Vir}(I) \subset \mathcal{A}(I)^G \subset (\mathcal{A}(I^c)^G)' \subset \mathrm{Vir}(I^c)' = \mathrm{Vir}(I)$ on that space, and local faithfulness of the representation lifts the equality to $\mathcal{A}$. We recomputed the Virasoro vacuum module in the charge-zero Fock space with exact rational arithmetic (Appendix B): its dimensions at levels $0, \dots, 9$ are $1, 0, 1, 1, 2, 2, 4, 4, 7, 8 = p(n) - p(n-1)$, as the character requires. Since Vir_1 is strongly additive, the reconstruction of Section 3 returns Vir_1 itself and not a dual net.

Corollary 4.7. *In $\mathcal{A}_{SU(2)_1}$ the exact fixed algebra of the Haar-jump noise $D^{SO(3)}$, of the semigroup $\tilde{D}$ built from finitely many fixed rotations that generate $SO(3)$ (two suffice), and of the pair of non-commuting axis noises H_z, H_x is $\mathrm{Vir}_1(0,\infty)$, a proper type III_1 subfactor of M.*

Properness: α is non-trivial on $\mathcal{A}$ and trivial on Vir_1.

Proposition 4.8 (The $\mathbb{Z}_2$ noise of the $U(1)$ current does not select the Virasoro net)**.** *Let $\mathcal{A}_{U(1)}$ be the net of the $U(1)$ current J, $\gamma : J \mapsto -J$, $F = \frac{1}{2}(\mathrm{id} + \gamma)$. Then $\mathrm{Vir}_1(I) \subsetneq \mathcal{A}_{U(1)}(I)^{\mathbb{Z}_2} \subsetneq \mathcal{A}_{U(1)}(I)$.*

Proof. At level 4 the even Fock space is spanned by $J_{-1}^4\Omega, J_{-3}J_{-1}\Omega, J_{-2}^2\Omega$, dimension 3, whereas the Virasoro vacuum module at level 4 has dimension 2, spanned by $L_{-4}\Omega = (J_{-3}J_{-1} + \frac{1}{2}J_{-2}^2)\Omega$ and $L_{-2}^2\Omega = (\frac{1}{4}J_{-1}^4 + J_{-3}J_{-1})\Omega$, with Gram matrix $\left(\begin{smallmatrix} 5 & 3 \\ 3 & 9/2 \end{smallmatrix}\right)$. The vector $u = \frac{1}{2}J_{-1}^4\Omega - J_{-3}J_{-1}\Omega + \frac{3}{4}J_{-2}^2\Omega$ is even, orthogonal to both, satisfies $L_m u = 0$ for $m = 1, \ldots, 4$, and has $\|u\|^2 = 27/2$ (Appendix B). If $\mathrm{Vir}_1(I) = \mathcal{A}(I)^{\mathbb{Z}_2}$, then by Reeh–Schlieder their vacuum-cyclic subspaces coincide: $F(\mathcal{A}(I))\Omega$ is dense in the even subspace and $\mathrm{Vir}_1(I)\Omega$ is dense in the Virasoro vacuum space; but u lies in the former and not the latter. The second inclusion is strict because γ is non-trivial. □

So the $\mathbb{Z}_2$ noise of the $U(1)$ current protects the stress tensor (and the whole even algebra) but does not select Vir_1; a noise that selects Vir_1 in that net must be specified differently, for instance by the conditional expectation onto Vir_1 itself. The $SU(2)$ construction has the advantage that the noise is specified by internal rotations, not by the target algebra.

Remark 4.9 (Fixed algebra, attracting state, correctable algebra). The three notions are of different types. The semigroups above preserve the vacuum state and act as the identity on the fixed algebra, so information in the fixed algebra is passively preserved and can be recovered by the identity map. A reset channel $E_t(A) = e^{-\gamma t}A + (1 - e^{-\gamma t})\mathrm{tr}(\rho_* A)\mathbf{1}$ has an attracting state ρ_* but trivial fixed algebra, and actively protected information is described by correctable algebras (Bény–Kempf–Kribs [4]). We do not identify a fixed algebra with an attracting state, and we do not claim that the full correctable structure of these semigroups equals their fixed algebra; exact fixed algebras are, moreover, discontinuous in the dynamics (Section 7).

5 Access: one observable generates the selected algebra

Let $\mathcal{A}$ be diffeomorphism covariant with $c > 0$, $M = \mathcal{A}(0, \infty)$, $B = \mathrm{Vir}_c(0, \infty)$. For $f \in C_c^\infty((0, \infty), \mathbb{R})$, $f \neq 0$, the *seed* is $O_f = \frac{2}{\pi}\arctan T(f) \in B$, a bounded self-adjoint observable. An *admissible* algebra is a unital von Neumann algebra $N \subset M$ with $\sigma_s(N) = N$ for all $s \in \mathbb{R}$ and $T_a(N) \subset N$ for $a \geq 0$. The *hull* $\mathrm{Hull}_{\sigma,T}(O_f)$ is the intersection of all admissible N containing O_f.

Theorem 5.1 (Generation)**.** $\mathrm{Hull}_{\sigma,T}(O_f) = \mathrm{Vir}_c(0, \infty)$ *for every nonzero* $f \in C_c^\infty((0, \infty), \mathbb{R})$.

Appendix A proves this for arbitrary nonzero f. The stronger modular-only result below applies to a specified bump class, not to all f.

Proposition 5.2 (Forward translations alone do not suffice)**.** *If* $\mathrm{supp} f \subset (a, b)$ *with* $a > 0$*, then the smallest algebra containing* O_f *and stable under* T_s*,* $s \geq 0$*, lies in* $\mathrm{Vir}_c(a, \infty) \subsetneq B$.

Proof. All forward translates of O_f lie in $\mathrm{Vir}_c(a, \infty)$, which is stable under forward translations. If $\mathrm{Vir}_c(a, \infty) = B$ then $\mathrm{Vir}_c(0, a) \subset B \cap B'$ by locality, which is trivial as B is a factor; but $\mathrm{Vir}_c(0, a)$ is not trivial. □

For the logarithmic bump class of Theorem 5.7, dilations alone suffice by Corollary 5.8. The unrestricted stress-seed question remains open. Foundation I gives a separate modular-only generation theorem for nonzero smeared current seeds [20].

A seed here denotes an ideal observable, not a grant of every function of it as an actual measurement or gate. An injective bounded function has the same spectral algebra as its generator; its implementation has a separate price. Foundation I distinguishes an effect, its spectral hull and the full physical instrument. HAFF-E supplies a concrete contrast: a partially polarized

memory produces an unsharp effect generating the same pointer algebra at every nonzero polarization, while its one-read error varies continuously [23]. The same distinction applies to O_f and to the ideal flows used below.

Example 5.3 (Exact fixedness does not freeze the modular flow)**.** In $M_2\bar{\otimes}M_2$ with state $\mathrm{diag}(.8,.2)\otimes\frac{1}{2}I$, let the noise be id$\otimes$ depolarisation and the observable $X\otimes 1$. The observable is fixed by the noise, but $\sigma_t(X\otimes 1) = (\cos(t\ln 4)X - \sin(t\ln 4)Y)\otimes 1$ generates the non-commutative algebra $M_2\otimes 1$. Exact invariance under the noise does not by itself prevent the modular flow from moving a fixed observable through a non-commutative algebra.

Theorem 5.4 (Access closure equals noise-selected algebra)**.** *In $\mathcal{A}_{SU(2)_1}$, let O_f be built from the Sugawara stress tensor. Then*

$$\mathrm{Hull}_{\sigma,T}(O_f) = \mathrm{Vir}_1(0,\infty) = M^{SO(3)} = \mathrm{Fix}(D^{SO(3)}) = \mathrm{Fix}(\tilde{D}).$$

Proof. The first equality is Theorem 5.1 applied inside M: the argument uses only unitaries $e^{iuT(h)} \in B \subset M$. The second is Theorem 4.6 (the half-line is a Möbius image of an interval), and the last two are Corollary 4.7. □

Two different structural definitions meet here: the algebra generated from one observable by dilations and forward translations, and the algebra left invariant by an explicit random-rotation noise. The equality is specific to this net and group; it fails for the $\mathbb{Z}_2$ noise of the $U(1)$ current (Proposition 4.8).

Remark 5.5 (Two different inclusions). The algebra selected by noise is $B = M^{SO(3)} = \mathrm{Vir}_1(0,\infty) \subset M = \mathcal{A}_{SU(2)_1}(0,\infty)$. By Lemma 4.2 there is an ω-preserving faithful normal conditional expectation of M onto B, so by Takesaki's theorem [24] $\sigma_s^M(B) = B$ for all $s\in\mathbb{R}$ and $\sigma^M|_B$ is the modular group of B in its own vacuum representation, so stability of B under σ in one direction holds trivially. What fails is cyclicity. For $x\in M$, $\int u(g)x\Omega\,dg = Q_{SO(3)}(x)\Omega$ because $u(g)\Omega=\Omega$; so the projection onto the $SO(3)$-invariant subspace $H^{SO(3)}$ maps $M\Omega$ onto $B\Omega$, and since $M\Omega$ is dense, $\overline{B\Omega} = H^{SO(3)}$, a proper subspace because the vacuum representation of $\mathcal{A}_{SU(2)_1}$ contains non-trivial isospin sectors. Ω is therefore not cyclic for B on the ambient Hilbert space, and $B\subset M$ is not a Cut in the sense of Definition 2.1 (which asks Ω to be cyclic and separating for the subalgebra; a proper subalgebra that is σ-invariant and has Ω cyclic would contradict Lemma 3.2). The Cuts of the geometric construction are the half-line subalgebras $N_a = \mathrm{Vir}_1(a,\infty)\subset B$, taken in the vacuum-cyclic representation of B (a factor, so the restriction to that subspace is faithful). The two structures meet in one technical model: noise fixes B, and B carries the net whose Cuts are the N_a. The noise selects B without producing its vacuum or modular dynamics. Theorem 5.7, applied in $H^{SO(3)}$, then constructs a standard Cut from a local seed and that modular dynamics, and reconstructs its translations. These need not be assumed as a separate generation permission. The underlying Virasoro carrier and state are still inherited from the ambient model.

5.1 A Cut generated by one past modular orbit

Here $M = \mathrm{Vir}_c(0,\infty)$ acts on its own vacuum-cyclic space. Write

$$N_{O_f}^- = W^*(\sigma_s(O_f) : s\le 0), \qquad H_\sigma(O_f) = W^*(\sigma_s(O_f) : s\in\mathbb{R}).$$

The first algebra is automatically half-sided invariant; it still needs cyclicity and properness before it is a Cut.

Lemma 5.6 (Smooth flow closure)**.** *For a unital von Neumann algebra N, the real compactly supported vector fields whose stress flows $e^{itT(v)}$ belong to N for every $t \in \mathbb{R}$ are closed under real linear combinations, under C^∞ limits with supports in a fixed compact set, and under the ordinary vector-field bracket $[v, w] = vw' - v'w$.*

Proof. These statements can be checked in any smooth coordinate chart covering the compact supports. Finite-energy vectors form a common core on which $T(v) + T(w) = T(v + w)$; essential self-adjointness and the Trotter formula give linear closure. Fixed-compact C^∞ convergence implies convergence of the circle fields in the Fourier norm $\sum_n |\hat{v}_n|(1 + |n|^{3/2})$. Carpi–Weiner's strong-resolvent result then gives strong convergence of their exponential unitaries [10, Thm. 4.4, Prop. 4.5].

For the bracket, put $\varphi_t = \mathrm{Exp}(tv)$. Up to scalar phases, $V(\varphi_t) = e^{itT(v)} \in N$. Geometric covariance gives

$$V(\varphi_t)e^{iuT(w)}V(\varphi_t)^* = e^{iu\lambda(t,w)}e^{iuT((\varphi_t)_* w)}, \qquad \lambda(t, w) \in \mathbb{R}.$$

The Schwarzian term contributes only the displayed scalar; projective phases do not change membership in a unital algebra. Thus $(\varphi_t)_* w$ has all its flows in N. Linear closure applies to $(w - (\varphi_t)_* w)/t$, and the smooth pushforward formula gives

$$\frac{d}{dt}(\varphi_t)_* w\Big|_{t=0} = v'w - vw' = -[v, w].$$

For sufficiently small $|t|$ all supports stay in one fixed compact set. The difference quotient converges in every C^k norm, so the preceding strong-resolvent closure proves the bracket claim. No commutator of unbounded operators on an unspecified domain is used. □

Theorem 5.7 (A single-orbit standard Cut)**.** *Choose $0 < a < b$, $\alpha = \log a$, $\beta = \log b$, and a real $g \in C_c^\infty(\mathbb{R})$ with support $[\alpha, \beta]$. Suppose $g > 0$ on (α, β), and g is strictly increasing and strictly decreasing on some interior neighborhoods of its left and right endpoints, respectively. Put $f(x) = xg(\log x)$ and $O_f = (2/\pi) \arctan T(f)$. Then*

$$\mathrm{Vir}_c(b, \infty) \subset N_{O_f}^- \subset \mathrm{Vir}_c(a, \infty) \subsetneq M. \tag{2}$$

The triple $(N_{O_f}^- \subset M, \Omega)$ is a proper standard +hsm inclusion. Its translations and strongly additive net are recovered from its modular operators.

Proof. In $u = \log x$, the vector field $f(x)\partial_x$ is $g(u)\partial_u$. A dilation by λ pushes g to $g(u - \log \lambda)$. Past modular times therefore give exactly $g_c(u) = g(u - c)$ with $c = -2\pi s \geq 0$. Injectivity of arctan gives all stress flows of g_c in $N = N_{O_f}^-$, as in Appendix A, Step 1. All these fields have support to the right of α, proving the upper inclusion in (2). For $r \leq 0$, $\sigma_r(N) \subset N$; equivalently, already obtained fields can be shifted further right by any $c \geq 0$.

Let $d = \beta - \alpha$ and choose $\varepsilon > 0$ smaller than both endpoint neighborhoods. Lemma 5.6 supplies

$$k_\varepsilon = [g_0, g_{d-\varepsilon}], \qquad \mathrm{supp}\, k_\varepsilon \subset [\beta - \varepsilon, \beta].$$

On the overlap, the first bump is nonincreasing and the second is nondecreasing, both positive. Hence $k_\varepsilon \geq 0$. Strict monotonicity need not mean a pointwise strict derivative. It does, however, give a positive integral: the nonconstant increasing shifted bump has a subinterval where its derivative is positive, and the other bump is positive there. Thus

$$m_\varepsilon = \int k_\varepsilon(u)\, du > 0, \qquad K_\varepsilon = k_\varepsilon / m_\varepsilon$$

is a unit-mass positive kernel whose stress flows and all right translates' flows belong to N.

For $h \in C_c^\infty((\beta,\infty),\mathbb{R})$, set

$$h_\varepsilon(u) = \int_{c\geq 0} h(c+\beta)K_\varepsilon(u-c)\,dc = \int K_\varepsilon(w)h(u-w+\beta)\,dw. \tag{3}$$

The second equality extends the integral without adding negative shifts: the support of $h(c+\beta)$ already has $c > 0$. If its distance from β is $\eta > 0$, then for $\varepsilon < \eta/2$ the supports of h_ε lie in a fixed compact subset of (β,∞). Since $w \in [\beta-\varepsilon,\beta]$ and the mass is one, for every $j \geq 0$,

$$\|h_\varepsilon^{(j)} - h^{(j)}\|_\infty \leq \varepsilon \|h^{(j+1)}\|_\infty.$$

The integral is a C^∞ limit of finite real Riemann sums of permitted right-translated kernels, all with fixed compact support. Lemma 5.6 first gives the flows of h_ε, then those of h, in N. Returning through $x = e^u$, these are all compact stress fields on (b,∞). They generate $\mathrm{Vir}_c(b,\infty)$, giving the lower inclusion.

Reeh–Schlieder makes Ω cyclic for the lower half-line algebra, hence for N; it is separating for $N \subset M$. Locality gives $\mathrm{Vir}_c(0,a) \subset M \cap N'$, so the relative commutant is also vacuum-cyclic. The upper half-line is proper: otherwise the nontrivial $\mathrm{Vir}_c(0,a)$ would lie in the center of the factor M. The past-orbit definition supplies half-sided invariance. Theorems 2.2 and 2.4 reconstruct translations and the net. □

Corollary 5.8 (Modular flow alone generates the full half-line)**.** *For the seeds in Theorem 5.7,*

$$H_\sigma(O_f) = \mathrm{Vir}_c(0,\infty).$$

Proof. The two-sided hull contains $N_{O_f}^-$, hence $\mathrm{Vir}_c(b,\infty)$, and is invariant under every dilation. It contains $\mathrm{Vir}_c(\lambda b,\infty)$ for every $\lambda > 0$. Every compact stress field on $(0,\infty)$ lies in one of these half-lines, so their generated algebra is M. □

For example, $g(u) = \exp[-1/((u-\alpha)(\beta-u))]$ inside its support and zero outside satisfies the hypotheses. Positivity can be reversed globally without changing the spectral algebra. The theorem is not an all-seed classification: sign-changing and more irregular endpoint behavior are not covered by the positive-kernel argument.

The lower inclusion is enough to prove a Cut, but does not identify $N_{O_f}^-$ with, or distinguish it from, an ordinary translated half-line. For $c > 1$, the standard reconstruction is strongly additive even though the original Virasoro net is not; one must not infer that the reconstructed full net equals the original net from the shared half-line algebra alone. For $c = 1$ the carrier is strongly additive, but whether this seed changes the isomorphism class of the reconstructed triple still requires an invariant.

Proposition 5.9 (Cyclicity supplies the remaining Cut conditions)**.** *For arbitrary nonzero f supported in a compact subset of (a,b), $a > 0$, in the same vacuum representation, cyclicity of $N_{O_f}^-$ implies that $N_{O_f}^- \subset M$ is a proper standard +hsm inclusion.*

Proof. The past orbit lies in $\mathrm{Vir}_c(a,\infty) \subsetneq M$, and $\mathrm{Vir}_c(0,a) \subset M \cap (N_{O_f}^-)'$. Locality and Reeh–Schlieder give standardness, while separatingness is inherited from M. □

Conjecture 5.10 (General seeds and the position of the Cut)**.** *(a) For every nonzero real compact stress seed on $(0,\infty)$, the past modular orbit is vacuum-cyclic. (b) Whenever the past orbit is cyclic, $N_{O_f}^-$ is not $\mathrm{Vir}_c(a',\infty)$ for any $a' > 0$.*

Part (a) is proved above for the endpoint-monotone positive log-bump class. Beyond that class, bracket closure remains available but a localized kernel of controlled nonzero mass has not been obtained in general. Part (b) is independent of cyclicity. Near the left support endpoint, finite bracket expressions built from right translates are multiples of the initial field. This germ restriction does not survive automatically through strong operator closure, so it cannot prove (b). A non-isomorphic geometry would require a further invariant of the full standard triple. The positive construction narrows the formation problem without assuming its remaining general conclusions.

6 Response of the selected algebra to excitation

Let $f \in C_c^\infty((a_0,\infty),\mathbb{R})$, $a_0 > 0$, $\rho = \mathrm{Exp} f$, $w = \rho''/\rho'$, and $0 \le a < a_0$. Put $N_a = \mathrm{Vir}_c(a,\infty)$.

Theorem 6.1 (Cut displacement)**.** *With $U = V(\rho)$, $\omega_\pm$ as in Section 2.3,*

$$D_a := S_{N_a}(\omega\|\omega_+) = S_{N_a}(\omega_-\|\omega) = \frac{c}{24}\int (x-a)\, w(x)^2\, dx.$$

Consequently

$$-\partial_a D_a = 2\pi E, \qquad D_0 - D_a = 2\pi a E, \qquad E = \frac{c}{48\pi}\int w^2\, dx = \int \langle T(x)\rangle_{\omega_-}\, dx,$$

where $\langle T\rangle_{\omega_-} = -\frac{c}{24\pi}\mathrm{Sch}(\rho)$ is the stress expectation in ω_-.

Proof. Since $\omega_+ \circ \mathrm{Ad}\, U = \omega$ and $\omega \circ \mathrm{Ad}\, U = \omega_-$, unitary covariance of the relative entropy gives $S(\omega\|\omega_+) = S(\omega_-\|\omega)$ on any algebra containing U, in particular N_a. Translation covariance moves the algebra N_a to N_0 and ρ to $\tilde\rho(y) = \rho(y+a) - a$, whose generating field $\tilde f(y) = f(y+a)$ is supported in $y > a_0 - a > 0$; (1) applied to $\tilde\rho$ gives $\frac{c}{24}\int y\,(\partial \log \tilde\rho')^2 dy = \frac{c}{24}\int (x-a) w^2 dx$. The same value is given by the exact identity for finite-amplitude local unitaries (not the linearised first law), $S_{N_a}(\omega_-\|\omega) = \langle U^*\Omega, -\log \Delta_{N_a} U^*\Omega\rangle = 2\pi \int (x-a)\langle T\rangle_{\omega_-} dx$, obtained from the transformation law of T and $\int (x-a)\mathrm{Sch}(\rho)\, dx = -\frac{1}{2}\int (x-a) w^2 dx$. Differentiating in a gives the remaining formulas. □

Corollary 6.2. *$G_a(f) = \partial_\theta^2 D_a(\mathrm{Exp}\,\theta f)|_{\theta=0} = \frac{c}{12}\int (x-a) f''(x)^2 dx$, which is > 0 for every non-zero test field f, and $G_0 - G_a = \frac{ca}{12}\int f''^2$.*

Proposition 6.3 (Noise does not reduce the response)**.** *Let $E : M \to B$ be a normal conditional expectation and $\iota : B \to M$ the inclusion. If two normal states φ, ψ on M factor through E ($\varphi = \varphi|_B \circ E$, $\psi = \psi|_B \circ E$), then $S_M(\varphi\|\psi) = S_B(\varphi|_B\|\psi|_B)$.*

Proof. Monotonicity under the restriction ι gives $S_B \le S_M$; monotonicity under the unital completely positive map E gives $S_M(\varphi\|\psi) \le S_B$. □

In $\mathcal{A}_{SU(2)_1}$, for $U \in \mathrm{Vir}_1$, the states ω and $\omega_\pm$ factor through $Q_{SO(3)}$: put $W_+ = U$, $W_- = U^*$, so that $\omega_\pm(x) = \omega(W_\pm^* x W_\pm)$ with $W_\pm \in M^{SO(3)}$; since $Q_{SO(3)}$ is $M^{SO(3)}$-bimodular and $\omega \circ Q_{SO(3)} = \omega$, $\omega_\pm(Q_{SO(3)}x) = \omega(Q_{SO(3)}(W_\pm^* x W_\pm)) = \omega_\pm(x)$; hence the information carried by an excitation of the selected algebra is the same in the full accessible algebra M as in Vir_1; the noise does not degrade it. This statement is about states generated inside the selected algebra, not about arbitrary information.

Remark 6.4 (What the identity is not). The identity is a statement about relative entropy on a family of algebras, and $-\partial_a D_a$ is a null-energy integral. It is not a Clausius relation for a thermodynamic entropy, and it does not produce an area or a curvature: moving a region in flat vacuum changes the algebra and its modular generator without creating curvature. The Hessian of the relative entropy corresponds to the Bogoliubov–Kubo–Mori metric, not to the symmetric-logarithmic-derivative quantum Fisher information. The structure is that of the half-sided modular inclusions used by Ceyhan and Faulkner [11] to relate shape derivatives of relative entropy to null energy, and null-plane algebras carry a copy of such a structure for each transverse position [12]; a transverse family with coupling is where area and focusing could enter, and is not treated here.

7 Limits of the results

Exact algebras and finite tasks. Exact fixed algebras and hulls are discontinuous in the dynamics. Two qubits with local Z-dephasing at rate κ and coupling $\epsilon X_1 X_2$ have the fixed algebra span$\{1, Z_1 Z_2\}$ for every $\epsilon \neq 0$, versus a four-dimensional diagonal algebra at $\epsilon = 0$. For $|\epsilon| < \kappa$, the slowest nonzero rate is $2(\kappa - \sqrt{\kappa^2 - \epsilon^2})$, so four slow modes persist on a window of order κ/ϵ^2. Their span is a rotated commutative algebra, but finite discrimination errors, not its exact fixedness, determine a reader's available tasks. HAFF-B treats such windows and the continuity of finite task risk [22].

Nor does the new generation theorem give a cheap continuum limit. For the normalized bump $g(u) = e^{4-1/(u(1-u))}$, the bracket mass at $\varepsilon = .05$ is about 1.1×10^{-31}. Dividing by that mass and taking spectral, Trotter and smooth limits are legitimate algebraic steps; they supply no bounded-control or finite-time certificate.

There is nevertheless a useful finite consequence of the constructive proof. Its scope is conditional on implementing the elementary flows, not on merely measuring the bounded seed.

Proposition 7.1 (A finite task consequence of flow generation)**.** *Take f, g, a, b as in Theorem 5.7, with $f_c(x) = xg(\log x - c)$. Suppose the elementary gates $e^{itT(f_c)}$, for finitely specified real t and $c \geq 0$, can be implemented to arbitrary specified error. For every compact stress field h supported in (b, ∞), every joint density operator ρ on the vacuum Hilbert space and a retained reference, and every $\delta > 0$, a finite word of such gates and their inverses approximates the channel of $e^{iT(h)}$ on ρ within trace distance δ. The statement holds uniformly over a specified finite list of joint states. If slot j is implemented within half-diamond error ϵ_j, the additional trace-distance error is at most $\sum_j \epsilon_j$.*

Proof. The proof of Theorem 5.7 uses conjugations, real linear combinations implemented by Trotter products, and fixed-compact smooth limits. For each constructed flow these give iterated strong limits of phase-adjusted finite words in the elementary flow unitaries. Scalar phases do not change the resulting channels. Their adjoints converge strongly too, since the limits are unitary and the inverse-time construction applies. On any finite set of test vectors, successive approximations can be chosen in the order in which these limits occur to produce one finite word with arbitrarily small vector error. On the separable vacuum Hilbert space a diagonal choice gives strong convergence of a sequence of words to $e^{iT(h)}$.

Strong convergence of uniformly bounded unitaries implies trace-norm convergence of conjugation on every trace-class joint state, first on finite-rank operators on the field and reference and then by trace-norm approximation. The same choice works for a finite list of states. For imperfect gates, telescoping through the same circuit and contractivity of channels gives the sum of slot errors, including the retained reference. □

This proposition is not uniform over all input states and gives no rate or optimal cost. Its targets are the stress flows constructed above, not every unitary or channel in the generated von Neumann algebra. Each finite approximation has a finite word length, gate precision, maximum modular shift and preparation budget that must be charged. $W^*(O_f)$ alone does not license its spectral functions as hardware. In a neutral current realization, that permission has a concrete implementation rather than an independent quadratic-control assumption.

Corollary 7.2 (Finite flows from local current control)**.** *Use the specified circle realization with neutral current Hamiltonian $H = L_0$, restricted at input to its $c = 1$ Virasoro vacuum subspace. Suppose signed local linear-current coupling, degenerate-helper rotations, record phase gates and calibrated clock preparation are available during coherent control intervals. Then the elementary-flow permission of Proposition 7.1 can be implemented by a finite positive autonomous program. For every fixed target there and $E < \infty$, its joint error can be made arbitrarily small, uniformly over inputs with $\mathrm{tr}(H\rho) \leq E$, retaining the old references, helper and original clock. The conclusion also holds on a specified finite list of joint states without a uniform energy bound.*

Proof. Q-RAIF C's current-stress compiler [1] selects finite current quadratures using full-period linear pulses, synthesizes their squares with a retained Pauli helper, and proves convergence of the current-factor cutoff to the Sugawara stress tensor. Its positive single-clock realization includes free propagation and both signs of each gate. Map the finitely many compact line smearings to the specified circle chart, accounting for scalar coordinate phases. On $P_M = 1_{[0,M]}(H)$, strong finite-word convergence becomes operator-norm convergence because P_M has finite rank. Approximate the finite set of prefix vectors when compiling the word and telescope the errors. For a joint mean-energy input the discarded proof tail costs at most $2\sqrt{E/(M+1)}$, independently of reference dimension. For a finite list of arbitrary joint states, their spectral tails instead tend to zero by trace-class convergence. The proof uses these projections only to bound errors, not as physical measurements or postselection. □

The physical choice has a bounded future test. Use the Fourier convention

$$f_n = (2\pi)^{-1} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta.$$

Take real compact f with $b = \mathrm{Im}\, f_2 \neq 0$, and $q_1 = (a_1 + a_1^*)/\sqrt{2}$. The vacuum flow satisfies

$$\langle e^{itT(f)}\Omega, q_1^2 e^{itT(f)}\Omega \rangle = \tfrac{1}{2} - bt + O(t^2).$$

Only the first-mode pair term contributes to this derivative. A later current Ramsey test of $e^{i\lambda q_1}$ therefore has $p'_+(0) = b\lambda^2 e^{-\lambda^2/4}/4 \neq 0$. A retained binary record selecting the two signs changes this later readout for sufficiently small nonzero t and implementation error. The test requires another paid linear pulse, helper and measurement. Intermediate current pulses can leave the Virasoro vacuum subspace; their endpoint leakage is included in the joint error. They are not protected by the always-on $SO(3)$ noise merely because the target stress flow is invariant. The carrier, coherent-control interval and growing precision budgets remain inputs; the finite program does not establish the exact Cut limit or its maintenance.

Foundation I gives a sharper, explicit bridge for finite pointer tasks: fresh partially polarized records approximate a sharp pointer instrument in half-diamond distance by their exact majority error, and errors add through finite adaptive circuits [20]. HAFF-A supplies a different finite achievement: actual calibrated fragment readouts certify distinct correlation-graph geometries

inside a stated GHZ protocol [21]. Neither result asserts that a finite apparatus has become an exact type-III Cut.

Sharing a carrier algebra does not fix its state: the vacuum and a local Virasoro excitation are both fixed by $D^{SO(3)}$ but have $D_0 > 0$. Conversely the two tensor-factor algebras intersect only in scalars while the correlated state $(1-\epsilon)(|00\rangle\langle 00| + |11\rangle\langle 11|)/2 + \epsilon I/4$ has Z disagreement $\epsilon/2$. Foundation II treats this distinction.

8 What the correspondence says about the independence assumption

A physical world can exist without a person looking at it. The harder question here concerns a complete description: does a perspective add an independently variable structure, or does it express relations already present in that world? The mathematical model isolates one structural part of this question. An accessible algebra and a state determine modular dynamics; a standard Cut and its vacuum determine a geometric net. Neither object is identified with experience.

Independence of variation, not ontological priority. Within the standard class, geometry and the modular triple determine each other. Fixing the net and vacuum fixes the modular flow of its half-lines; fixing a standard Cut returns the net. A geometric description and this structural proxy for a perspective cannot then be varied as two independent pieces of data. The equivalence is symmetric. A net with a vacuum remains a legitimate starting specification; it contains its half-line Cuts without waiting for an observer. Conversely, the standard triple is enough to reconstruct that description. Choosing either notation first settles no question about what exists first.

The choice of usable structure is a different question. Theorem 5.7 makes progress on it: once a local bump seed, carrier and modular dynamics are given, cyclicity, properness and standardness are consequences, not further choices. A reconstructed translation group is obtained without an independent translation permission. The result no longer leaves every Cut as an external input. It still inherits the state and carrier that define the modular flow. Noise selection also acts on a specified carrier: two internal rotations select the Virasoro subnet, but do not create its vacuum dynamics from nothing.

From a selected boundary to a maintained perspective. The seed-generated Cut supplies a boundary within which local operations can be reconstructed. A physical reader must additionally acquire records, retain them, choose operations and survive the disturbance its operations cause. The finite instruments of HAFF-E and Foundation I show how this extra layer can have explicit errors and resources. The geometry certification of HAFF-A then shows that changing actual record protocols can change what geometry is certified under the same dictionary. Those finite graphs and the chiral modular net are different geometric objects; no common continuum limit is asserted between them.

This distinction supports a stronger mechanism than treating perspective as a passive label. A record can change a subsequent coupling and thereby change the state or source, not merely the description. The subject line of the series asks when a retained history makes such changes useful and when a system must distinguish world drift from drift of its own interface. The present Cut theorem supplies an ideal structural boundary to that programme; it does not require a human reader, and it does not replace the programme with a net-theoretic definition of consciousness.

Conjecture 8.1 (Maintained modular boundaries)**.** *In a suitable local carrier and stationary reference state, a physical record-and-control process can maintain a hierarchy of finite task permissions whose modular covariance and nesting errors tend to zero as the charged control, memory and preparation resources increase, and whose ideal limit is a standard seed-generated Cut. Changing the retained control history can change the maintained boundary and hence subsequent response and access.*

The single-orbit construction identifies a nonempty ideal limit. Corollary 7.2 implements the finite words from specified local current controls, uniformly on bounded-energy inputs with retained references. The remaining task is to form those primitive permissions and sustain the control process, with explicit preparation, transport and maintenance costs on the same history. The claim would fail for a proposed process if its covariance error stayed bounded away from zero at increasing budgets, if disturbance destroyed standardness in every admissible stationary limit, or if different histories only relabelled an unchanged instrument. A family of task-level bounds, rather than algebra dimension alone, is the relevant evidence.

Experience and competing readings. The explanatory-gap argument [8] is not refuted by exchanging a geometric description for an algebraic one. A reductionist can read the Cut as a convenient organization of observables; a relational reading can treat the same dependencies as evidence that a perspective and its usable world are jointly constituted. Both readings must accept the same correspondence and finite interventions. Neither obtains experience from it. Thermal time [9] motivates the modular proxy, but state-relative modular evolution exists whether or not the system reports anything.

A stronger claim about first-person organization requires additional distinctions: private history versus public calibration, causal self-reading versus a model supplied from outside, and reporting behavior versus experience. Consciousness III develops that bridge as its own problem. Here the strongest supported conclusion is structural: the boundary, its state response and its geometric reconstruction are constrained together, and one such boundary can be generated from local seed history in a given carrier. That is a restriction on independent specification, not a proof that the universe needs an observer.

Null-plane copies have separate transverse source and coupling requirements [12, 11]; Consciousness II and HAFF-D address those physical interfaces.

9 Open problems

1. **Physical formation.** Derive the carrier, stationary reference and availability of the primitive local current controls from a common process, and test Conjecture 8.1. The seed-generated standard Cut is constructed in the specified model; primitive modular dynamics and a finite maintenance protocol are not.
2. **General seeds and inequivalent Cuts.** Conjecture 5.10 outside the positive log-bump cyclicity theorem; whether a seed-generated standard triple differs from all ordinary translated half-line triples. Modular-only generation is proved for that bump class, not for every nonzero stress seed.
3. **Finite resources.** Determine explicit gate costs and convergence rates for Corollary 7.2. It already supplies uniform approximability on bounded-energy inputs with arbitrary retained references; it does not give an optimal budget or a maintained Cut limit.

4. **Transverse families and gravity.** Coupling between null-plane copies, area and focusing; the selection of the algebra family and the physical source enter here. An invisible source is not thereby absent; changing modular data and changing actual stress are different tests. For ball families in higher dimensions Consciousness II identifies the variation of area minus $(d-1)/R$ times volume as minus the weighted linearized Einstein charge. On a conformal net in Minkowski space, its supplied MC_η consistency law with the CHM bulk modular energy, after the declared boundary-charge subtraction and under that paper's source and functional assumptions, is equivalent to the linearized Einstein equation [19]; null-plane couplings remain open.

5. **Selection.** Why this noise group; how unrecorded transformations are aligned or reduced and at what cost; the imprint of the discarded group in the sectors of the selected algebra (for SU(2) level one: the subcategory of Virasoro sectors of lowest weight j^2 contained in the vacuum representation).

6. **Unbounded generators.** Which non-multiplicative semigroups are compatible with a strict scaling covariance $E_t\sigma_s = \sigma_s E_{e^{2\pi s}t}$ is unclassified: it is known that such a semigroup preserves the vacuum, that it is trivial if it commutes with σ, and that it is zero if its generator is bounded.

A Proof of Theorem 5.1

Write $B = \mathrm{Vir}_c(0,\infty)$, $\mathrm{Hull} = \mathrm{Hull}_{\sigma,T}(O_f)$, and for a von Neumann algebra $N \subset M$ $\mathcal{V}_N = \{h \in C_c^\infty((0,\infty),\mathbb{R}) : e^{iuT(h)} \in N\ \forall u \in \mathbb{R}\}$.

Step 0: Hull $\subset B$. B is admissible: $\sigma_s = \mathrm{Ad}\,V(\delta_{e^{-2\pi s}})$ preserves the supports $(0,\infty)$ and the generators $T(h)$ go to $T(\delta_* h)$; T_a, $a \geq 0$, moves supports into $(a,\infty) \subset (0,\infty)$; and $O_f \in B$.

Step 1: spectral projections. $O_f = \psi(T(f))$ with the injective Borel function $\psi = \frac{2}{\pi}\arctan$. Hence ψ^{-1} is Borel, every bounded Borel function of $T(f)$ is a bounded Borel function of O_f, and so $e^{iuT(f)} \in W^*(O_f) \subset N$ for every admissible $N \ni O_f$. In particular $f \in \mathcal{V}_N$; $\mathcal{V}_N$ is closed under multiplication by real scalars since u ranges over $\mathbb{R}$.

Step 2: affine images. For $\varphi(x) = \lambda x + b$, $\lambda > 0$, $b \geq 0$, the Schwarzian vanishes, so $V(\varphi)T(h)V(\varphi)^* = T(\varphi_* h)$ with $(\varphi_* h)(y) = \lambda h((y-b)/\lambda)$. If $h \in \mathcal{V}_N$ then $e^{iuT(\varphi_* h)} = T_b\big(\sigma_s(e^{iuT(h)})\big)$ with $e^{-2\pi s} = \lambda$ belongs to N, because N is stable under σ and under T_b. So $\varphi_* h \in \mathcal{V}_N$.

Step 3: linear and limit closure. If $h_1, h_2 \in \mathcal{V}_N$ then $h_1 + h_2 \in \mathcal{V}_N$: $T(h_1) + T(h_2)$ equals $T(h_1 + h_2)$ on the finite-energy vectors, a common core on which $T(h_1 + h_2)$ is essentially self-adjoint [10, Thm. 4.4], so the Trotter product formula $e^{iu(A+B)} = \text{s-}\lim(e^{iuA/n}e^{iuB/n})^n$ gives $e^{iuT(h_1+h_2)} \in N$, N being strongly closed. If $h_n \in \mathcal{V}_N$ have supports in a fixed compact subset of $(0,\infty)$ and $h_n \to h$ in C^∞ (hence in $\|\cdot\|_{3/2}$), then $e^{iuT(h_n)} \to e^{iuT(h)}$ strongly [10, Prop. 4.5], and $h \in \mathcal{V}_N$.

Step 4: the case $m = \int f \neq 0$. Let $h \in C_c^\infty((0,\infty),\mathbb{R})$ and put $f_{\varepsilon,b}(y) = \varepsilon f((y-b)/\varepsilon)$, $k_{\varepsilon,b} = f_{\varepsilon,b}/(\varepsilon^2 m)$, so $\int k_{\varepsilon,b} = 1$ and $k_{\varepsilon,b} \in \mathcal{V}_N$ by Steps 1–3. Then

$$h_\varepsilon(y) := \int_{b\geq 0} h(b) k_{\varepsilon,b}(y)\, db = \frac{1}{m}\int f(w)\, h(y-\varepsilon w)\, dw \longrightarrow h(y)$$

in C^∞, with supports in a fixed compact subset of $(0,\infty)$ for small ε. Only $b \geq 0$ is used: h has compact positive support, and $h(y-\varepsilon w) = 0$ whenever $b = y - \varepsilon w < 0$. Riemann sums of the integral are finite real combinations of the k_{ε,b_i}, which lie in $\mathcal{V}_N$ and converge to h_ε in C^∞ with fixed compact support, so $h_\varepsilon \in \mathcal{V}_N$ by Step 3, and then $h \in \mathcal{V}_N$. Hence $\mathcal{V}_N = C_c^\infty((0,\infty),\mathbb{R})$.

Step 5: the case $\int f = 0$. Let $\varphi_s = \mathrm{Exp}(sf)$. By Step 1, $e^{isT(f)} \in N$, and $e^{isT(f)} = V(\varphi_s)$ up to a phase, so conjugation by it maps $\mathcal{V}_N$ to itself via $h \mapsto (\varphi_s)_* h$. (For a general diffeomorphism the conjugation sends $T(h)$ to $T(\varphi_* h)$ plus a real multiple of the identity, the Schwarzian term; the multiple exponentiates to a phase, which does not affect membership in N since $N \ni 1$.) With $f_\delta(y) = f(y-\delta)$, $\delta \geq 0$, we get $(\varphi_s)_* f_\delta \in \mathcal{V}_N$. Since $\frac{d}{ds}(\varphi_s)_* g = -[f,g]$ with $[X,Y] = XY' - X'Y$ and $\int [f, f_\delta] = 2\int f f_\delta'$,

$$\frac{d}{ds}\Big|_0 \int (\varphi_s)_* f_\delta\, dy = -2\int f(y) f'(y-\delta)\, dy = 2C'(\delta), \qquad C(\delta) = \int f(y) f(y-\delta)\, dy.$$

C is even, smooth, compactly supported and $C(0) = \|f\|^2 > 0$, so $C'(\delta_0) \neq 0$ for some $\delta_0 > 0$. For small $s \neq 0$ the field $g = (\varphi_s)_* f_{\delta_0}$ has $\int g \neq 0$, compact support in $(0,\infty)$ and $g \in \mathcal{V}_N$; apply Step 4 to g.

Step 6. All $e^{iuT(h)}$, $h \in C_c^\infty((0,\infty),\mathbb{R})$, lie in N, so $N \supset \{T(h)\}'' = B$. With Step 0, Hull $= B$. $\square$

B The Virasoro vacuum module in Fock space

In the Fock space of the $U(1)$ current with $[J_m, J_n] = m\delta_{m+n,0}$, $J_n\Omega = 0$ for $n \geq 0$, take $L_n = \frac{1}{2}\sum_k :J_{n-k}J_k:$. We built the descendants $L_{-n_1}\cdots L_{-n_k}\Omega$ ($n_i \geq 2$) at each level with exact rational arithmetic and computed their rank, and compared with the number of partitions into an even number of parts (the $\mathbb{Z}_2$-even Fock space).

level n	0	1	2	3	4	5	6	7	8	9
dim Virasoro vacuum module	1	0	1	1	2	2	4	4	7	8
$p(n) - p(n-1)$	1	0	1	1	2	2	4	4	7	8
dim even Fock space	1	0	1	1	3	3	6	7	12	14

The first level where the even space is strictly larger is 4. At level 4, $\langle L_{-4}\Omega, L_{-4}\Omega\rangle = 5 = \frac{c}{12}(4^3-4)$ with $c=1$, the Gram matrix on $\{L_{-4}\Omega, L_{-2}^2\Omega\}$ is $\left(\begin{smallmatrix} 5 & 3 \\ 3 & 9/2 \end{smallmatrix}\right)$ with determinant $27/2$, and $u = \frac{1}{2}J_{-1}^4\Omega - J_{-3}J_{-1}\Omega + \frac{3}{4}J_{-2}^2\Omega$ satisfies $L_1 u = L_2 u = L_3 u = L_4 u = 0$, $\langle u, L_{-4}\Omega\rangle = \langle u, L_{-2}^2\Omega\rangle = 0$, $\|u\|^2 = \frac{1}{4}\cdot 24 + 3 + \frac{9}{16}\cdot 8 = \frac{27}{2}$; the three vectors have rank 3. The generating function of the even dimensions agrees through order 23 with $\sum_{p\geq 0}(t^{4p^2} - t^{(2p+1)^2})\prod_n (1-t^n)^{-1}$, and the $SU(2)_1$ vacuum character identity $\sum_m z^m t^{m^2} P(t) = \sum_j \chi_j(z)(t^{j^2} - t^{(j+1)^2})P(t)$, $P(t) = \prod(1-t^n)^{-1}$, holds coefficient by coefficient for $|m| \leq 5$ through order 23. These finite checks are consistent with the decomposition of the even Fock space as $\bigoplus_p L(1, 4p^2)$ and with the character identity behind Theorem 4.6, but a check through finitely many levels does not prove an infinite decomposition; only the level-4 computation is used in Proposition 4.8, and the identity of characters used for Theorem 4.6 is Rehren's.

C Numerical checks

Cut displacement. For $f(x) = e^{4-1/(z(1-z))}(1 + 0.7(z - \frac{1}{2}))$, $z = x - 2$, supported in $(2, 3)$, $c = 1$, $\rho = \mathrm{Exp}(\theta f)$, the integrals of Theorem 6.1 were evaluated by quadrature after integrating the flow and its first three variational derivatives. For $\theta = 0.05$:

a	0	1	1.5	1.9
D_a/D_0	1.000	0.604	0.406	0.2476
$D_0 - D_a$	0	4.5641e-2	6.8461e-2	8.6718e-2
$2\pi aE$	0	4.5641e-2	6.8461e-2	8.6718e-2

and both $-\partial_a D_a$ and $2\pi \int \langle T \rangle dx$ are 4.5640958×10^{-2}. At 2001 and 4001 spatial points the displayed values agree to the shown precision. This is a quadrature diagnostic, not an interval-certified error bound. For $\theta = 0.2$, $D_a/D_0 = 1, 0.593, 0.390, 0.227$ at the same a.

Single-orbit closure. The analytic proof uses the allowed kernel integral (3), without negative recentering. For $g(u) = e^{4-1/(u(1-u))}$ on $(0, 1)$, numerical quadrature checks the kernel's non-negative sign and positive mass, and convergence in the first three derivative norms for targets supported strictly to the right of 1. A separate geometric-flow check evaluates the pushforward difference quotient of Lemma 5.6. These diagnostics check signs and a finite selection of derivatives; the proof of C^∞ and operator closure is the analytic argument, not the grid. Small kernel mass is also recorded, since normalization can be extremely costly and cannot certify finite-resource generation.

Finite-dimensional checks. The identities of Theorems 4.4 and 4.5 were checked in the spin representation $[1, 1, 2]$ of $SO(3)$ on M_{11}: the fixed algebra is 5-dimensional for the two-axis noise, the axis noises do not commute ($\|[Q_z, Q_x]\|_F = 1.90$), the dissipative identity of Theorem 4.5 holds for a fixed complex random probe (both sides 363.2646789268), and for $L = Q_x + Q_z - 2\,\mathrm{id}$, $\|e^{tL} - Q_{SO(3)}\|$ falls from 0.60653 at $t = 1$ to 9.4×10^{-14} at $t = 60$. The two specified irrational rotations also have a five-dimensional fixed algebra in this representation. These are finite-dimensional analogues, not statements about the net.

D Boundaries of shell and scale rules

A shell containing a full norm ball generates the ambient algebra and cannot select a proper Cut; a unit-sphere version has the same defect. Modular covariance must respect the antilinearity identity $J\Delta^{is}J = \Delta^{is}$. The vacuum modular generator has zero variance, while a truncated one-sided variance can diverge and have a regularization-dependent finite part. Approximate innerness does not imply innerness of an automorphism. Faithfulness gives separation in the stated representation, not cyclicity. These boundaries are separate from the local stress-seed proof. The complete earlier-route analysis is retained in the supplementary source record.

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